

# **mars\_moc**

Processed on: 2026-04-28 16:18:00

ASP version: 3.7.0-alpha

asp\_plot version: 2.0.0

## **DEM Summary**

| Property            | Value               |
|---------------------|---------------------|
| DEM File            | output-DEM.tif      |
| Dimensions (px)     | 1201 x 4145         |
| GSD (m)             | 4.44                |
| CRS                 | EPSG:None           |
| Nodata (%)          | 45.2                |
| Elevation Range (m) | -10326.0 to -9973.7 |

# Processing Parameters

## Runtime Summary

| Step          | Runtime               |
|---------------|-----------------------|
| Bundle Adjust | N/A                   |
| Stereo        | 0 hours and 3 minutes |
| point2dem     | 0 hours and 0 minutes |

### Bundle Adjust Command:

Bundle adjustment not run

### Stereo Command:

```
stereo --stereo-algorithm asp_mgm --subpixel-mode 9 --alignment-method none E0201461.map.cub M0100115.map.cub  
out_stereo_proj/output --corr-seed-mode 1 --sgm-collar-size 256 --compute-point-cloud-center-only --threads 8
```

### point2dem Command:

```
point2dem --auto-proj-center --errorimage out_stereo_proj/output-PC.tif
```

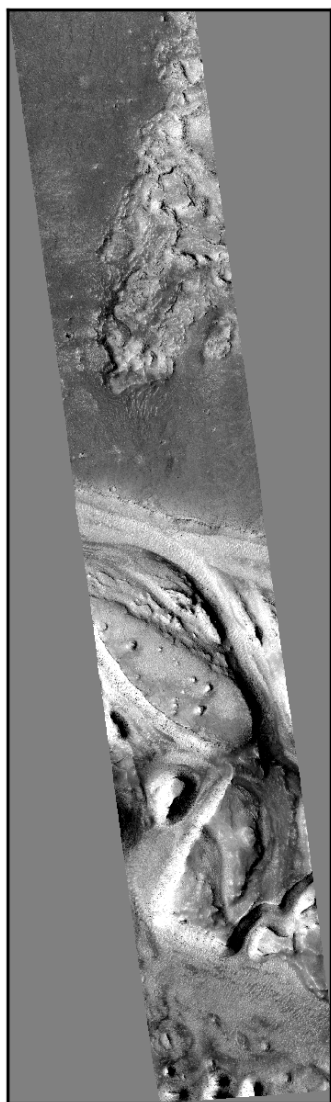
### Report Generation Command:

```
asp_report --directory /Users/ben/Desktop/asp-plot-examples/mars_moc/ --stereo_directory out_stereo_proj/ --add_basemap  
False --altimetry_csv /Users/ben/Desktop/asp-plot-  
examples/mars_moc/438115_mola/MolaPEDR_34N34N_142E142E_20260319T204814563_pts_csv.csv --plot_geometry False --subset_km  
0.5 --reuse_selections /Users/ben/Desktop/uw-github/asp_plot/reports/MOC-asp-report_figure_selections.yml  
--report_filename /Users/ben/Desktop/uw-github/asp_plot/reports/MOC_mapproj-asp-report.pdf
```

## Input Scenes

### Input Scenes Map-projected Scenes

Left  
output-L\_sub.tif



Right  
output-R\_sub.tif

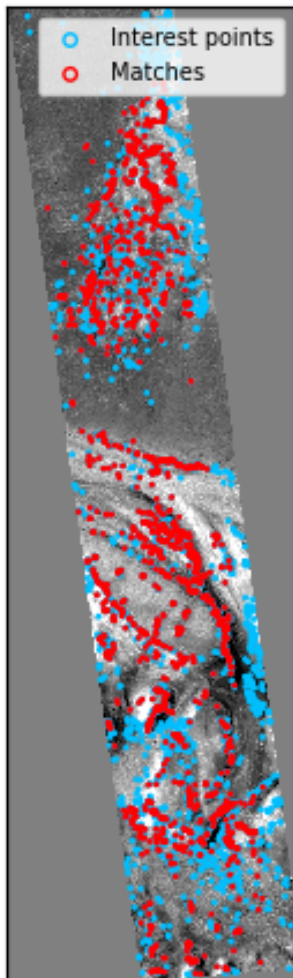


Figure 1: Left and right input scenes used for stereo processing. Non-mapprojected scenes are shown after ASP's alignment step (e.g., *affineepipolar*), which rotates images to create horizontal epipolar lines for correlation. Mapprojected scenes have been orthorectified with RPCs against a reference DEM to roughly align the two images prior to correlation, which reduces the disparity search range; they are displayed here in their map-projected orientation.

## Match Points

### Stereo Match Points

Left (3,164 ip, 1,331 matches)



Right (3,419 ip, 1,331 matches)

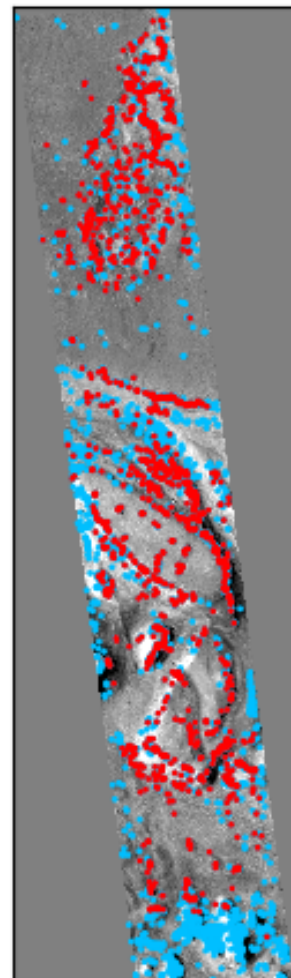


Figure 2: Interest point matches between left and right images (red), overlaid on the raw interest points detected on each image individually (blue, when the .vwip files are present in the stereo directory). These are produced by stereo\_corr during its initial interest point matching step, which is used to set the search windows for subsequent dense correlation (not the dense correlation matches themselves). Comparing the raw interest points against the matches shows whether sparse matching traces back to poor matching or to areas with no detected interest points at all (e.g., limited texture, clouds, water). Some ASP configurations only write a .vwip file for the left image (the right image's interest points are detected on a temporary aligned image and not saved), in which case only matches are shown on the right.

## Disparity

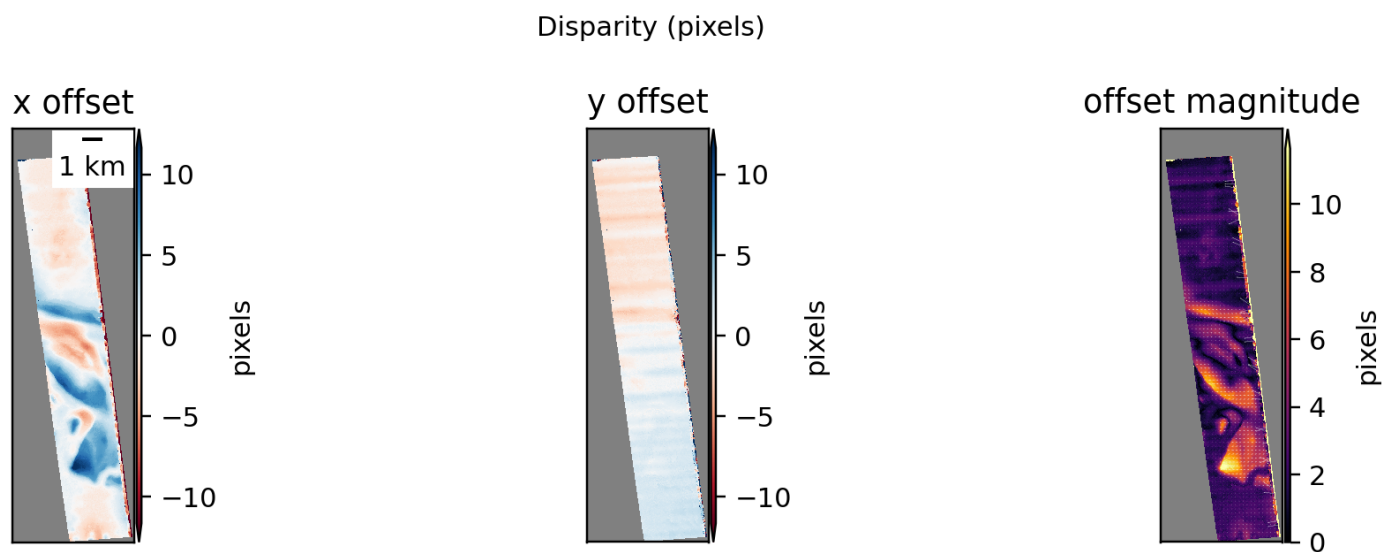


Figure 3: Horizontal and vertical disparity maps in pixels with quiver overlay.

## DEM Results

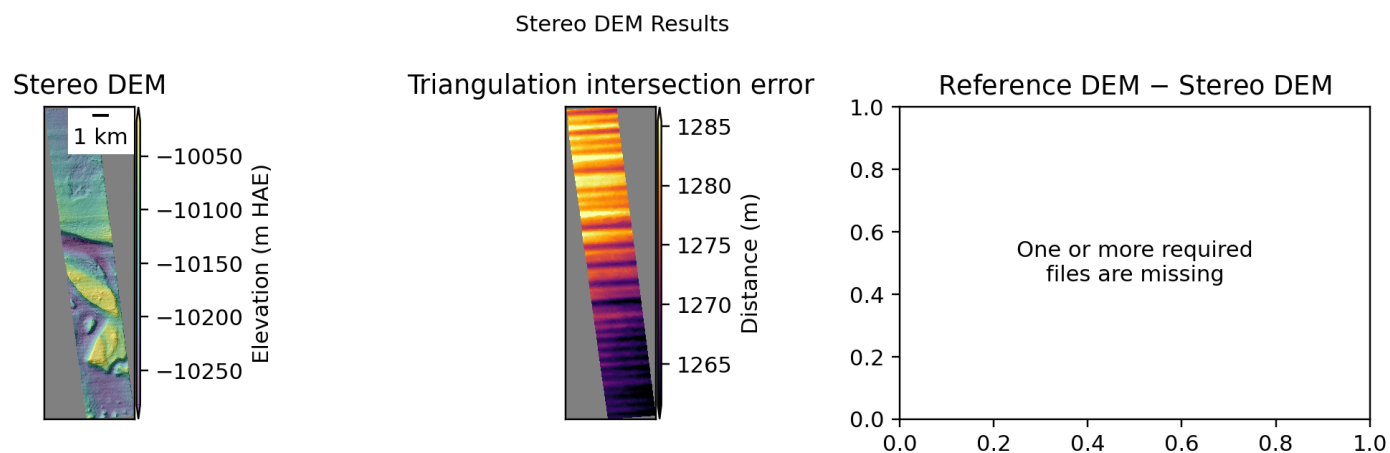


Figure 4: Output DEM with intersection error map and difference relative to the reference DEM used in processing.

## Detailed Hillshade

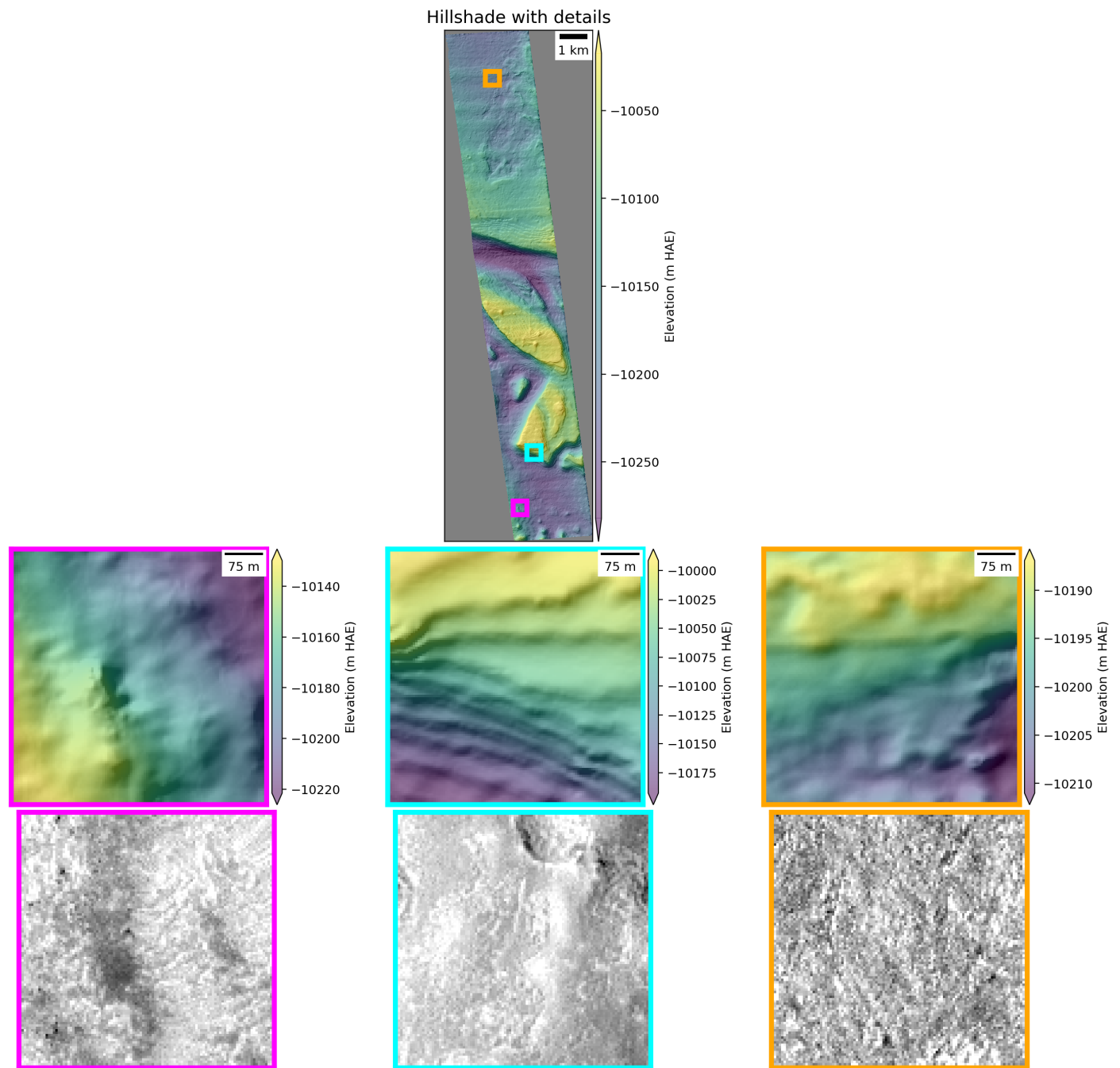


Figure 5: DEM hillshade. If the intersection error is available, zoomed subsets selected from low, medium, and high (left to right) uncertainty areas are displayed in the second row. If the mapprojected image is available, corresponding ortho image subsets are displayed in the bottom row.

## MOLA Altimetry Map

MOLA vs DEM

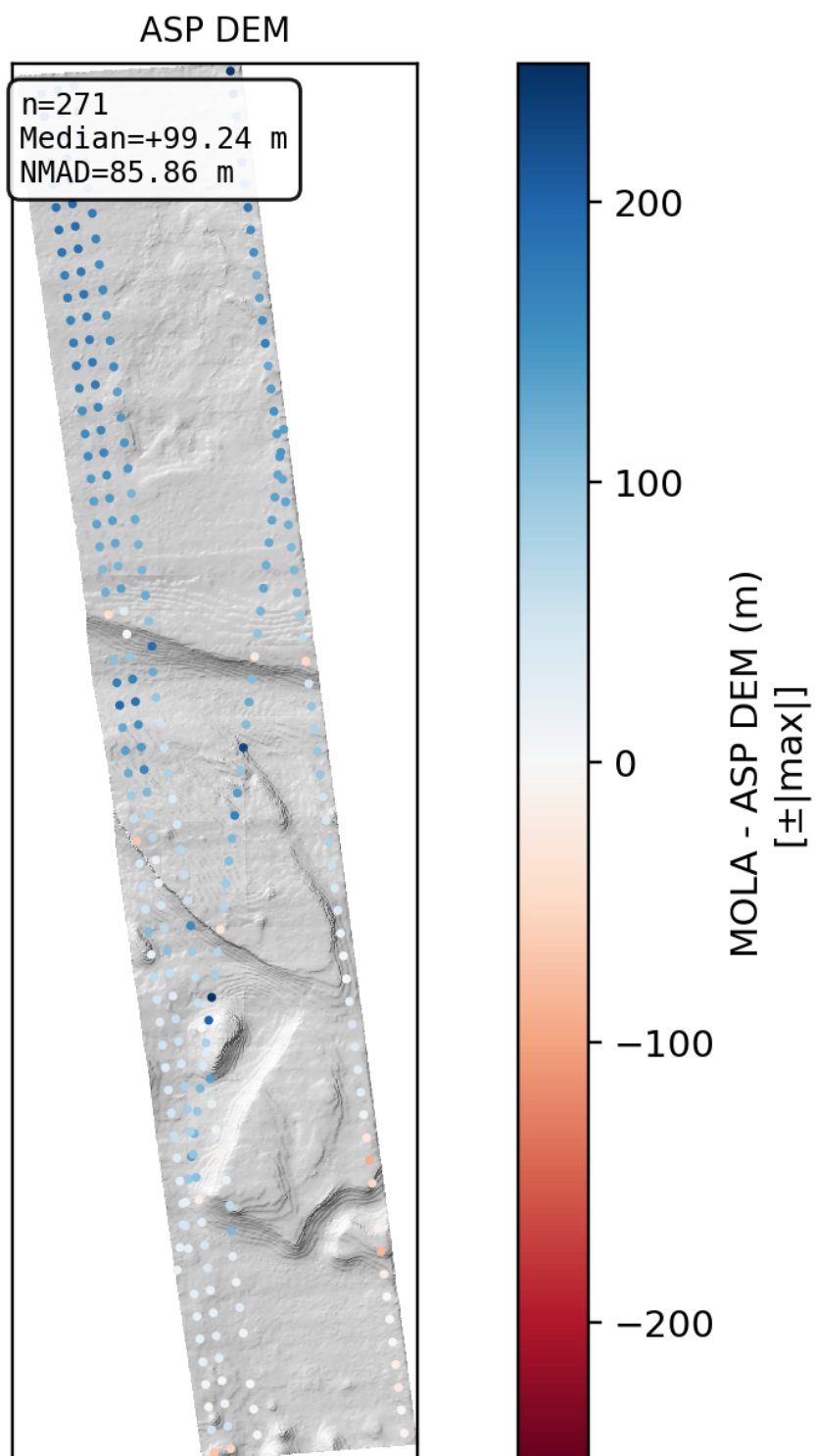


Figure 6: MOLA elevation differences vs. ASP DEM.



## MOLA Altimetry Histogram

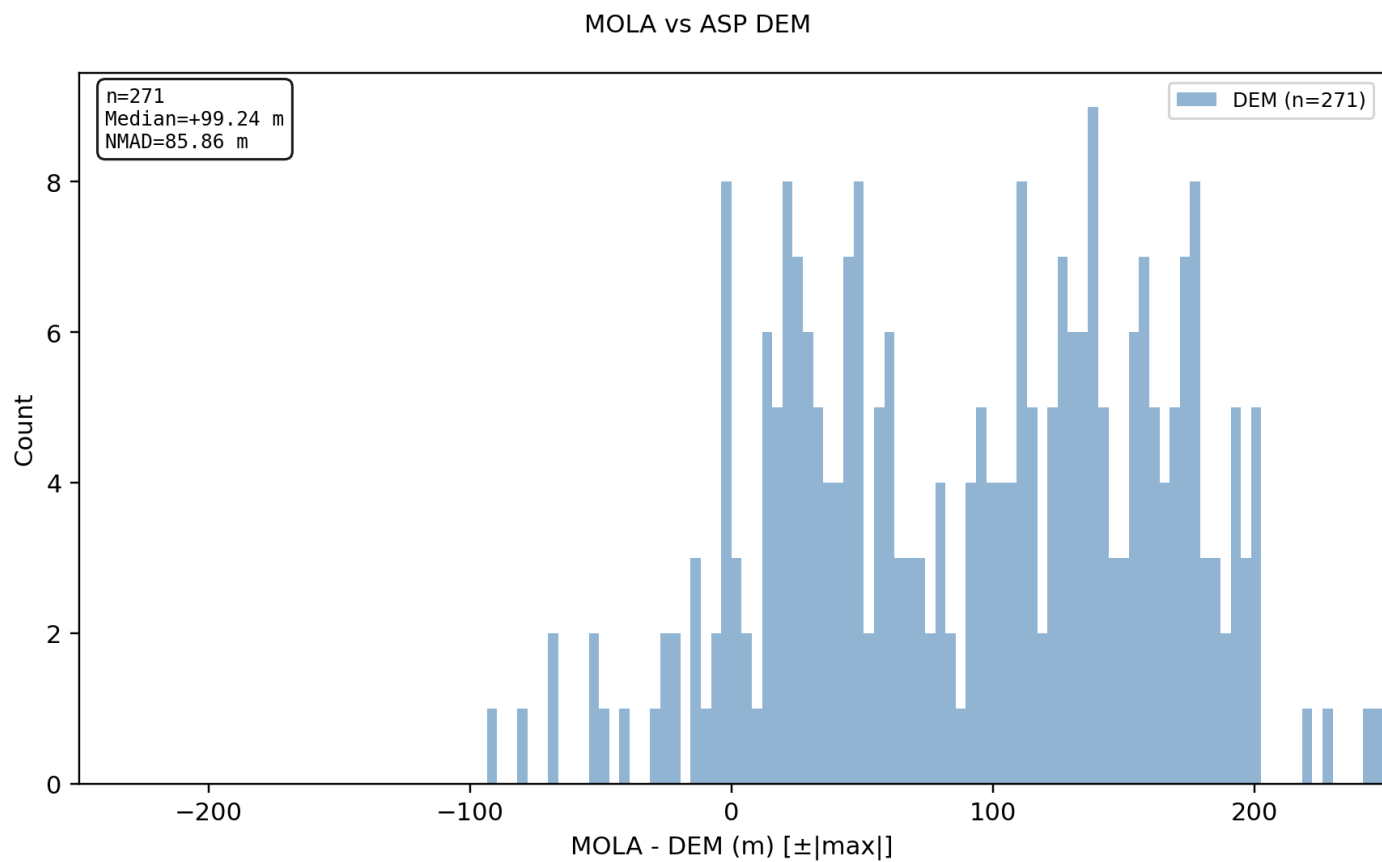


Figure 7: Distribution of elevation differences between MOLA and ASP DEM.

## DEM Alignment with MOLA

### Alignment Parameters

| Parameter                 | Value |
|---------------------------|-------|
| max_displacement          | 500   |
| minimum_points            | 20    |
| min_translation_threshold | 0.1   |
| improvement_threshold_pct | 5.0   |

### Alignment Statistics (m)

| p16_beg | p50_beg | p84_beg | p16_end | p50_end | p84_end | N_shift | E_shift | D_shift | T   |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|-----|
| 25.8    | 103     | 173     | 23.4    | 60.4    | 98.1    | 34.1    | 29.1    | 94.4    | 104 |

ASP's `pc_align` estimates a rigid 3D translation that minimizes the height residuals between the ASP DEM and the MOLA planetary radii. The CSV is passed as the reference cloud with `--csv-format '1:lon 2:lat 3:radius_m'`, and `--datum` is set to `D_MARS` or `D_MOON` to match the ASP DEM. The resulting translation is applied to the DEM directly (geotransform + pixel-value shift, no resampling) to produce the aligned DEM.

Alignment Parameters (above):

- `max_displacement`: `pc_align` upper bound on the translation magnitude (m).
- `minimum_points`: minimum MOLA points that overlap the DEM; below this the alignment is skipped.
- `min_translation_threshold`: minimum translation magnitude (as a fraction of the DEM GSD) required to write out an aligned DEM.
- `improvement_threshold_pct`: minimum percentage reduction in p50 required to keep the aligned DEM on disk; below this, the aligned DEM is removed.

Alignment Statistics (above, in meters):

- `p16_beg` / `p50_beg` / `p84_beg`: 16th / 50th / 84th percentile of the DEM-vs-MOLA absolute height residuals before alignment.
- `p16_end` / `p50_end` / `p84_end`: same percentiles after alignment.
- `N_shift` / `E_shift` / `D_shift`: north / east / down components of the applied translation vector.
- `|T|`: magnitude of the translation vector.

**p50 improved from 103.22 m -> 60.42 m (41.5% reduction). Aligned DEM written to `/Users/ben/Desktop/asp-plot-examples/mars_moc/out_stereo_proj/output-DEM_pc_align_translated.tif`.**

## MOLA Altimetry Map (Aligned DEM)

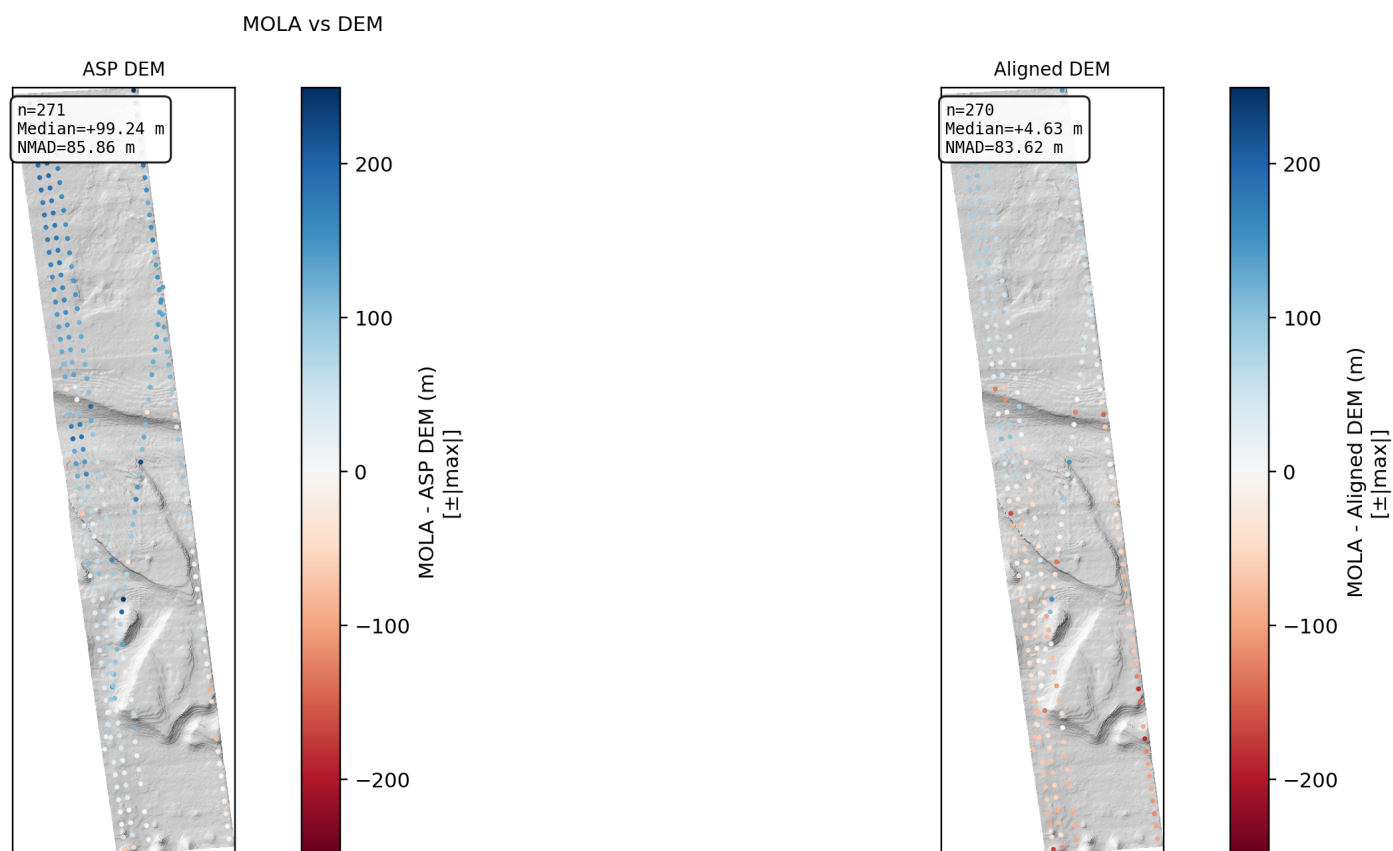


Figure 9: Pre- (left) and post-alignment (right) map views of MOLA elevation differences. The aligned-DEM hillshade is used as the backdrop for both panels.

## MOLA Altimetry Histogram (Aligned DEM)

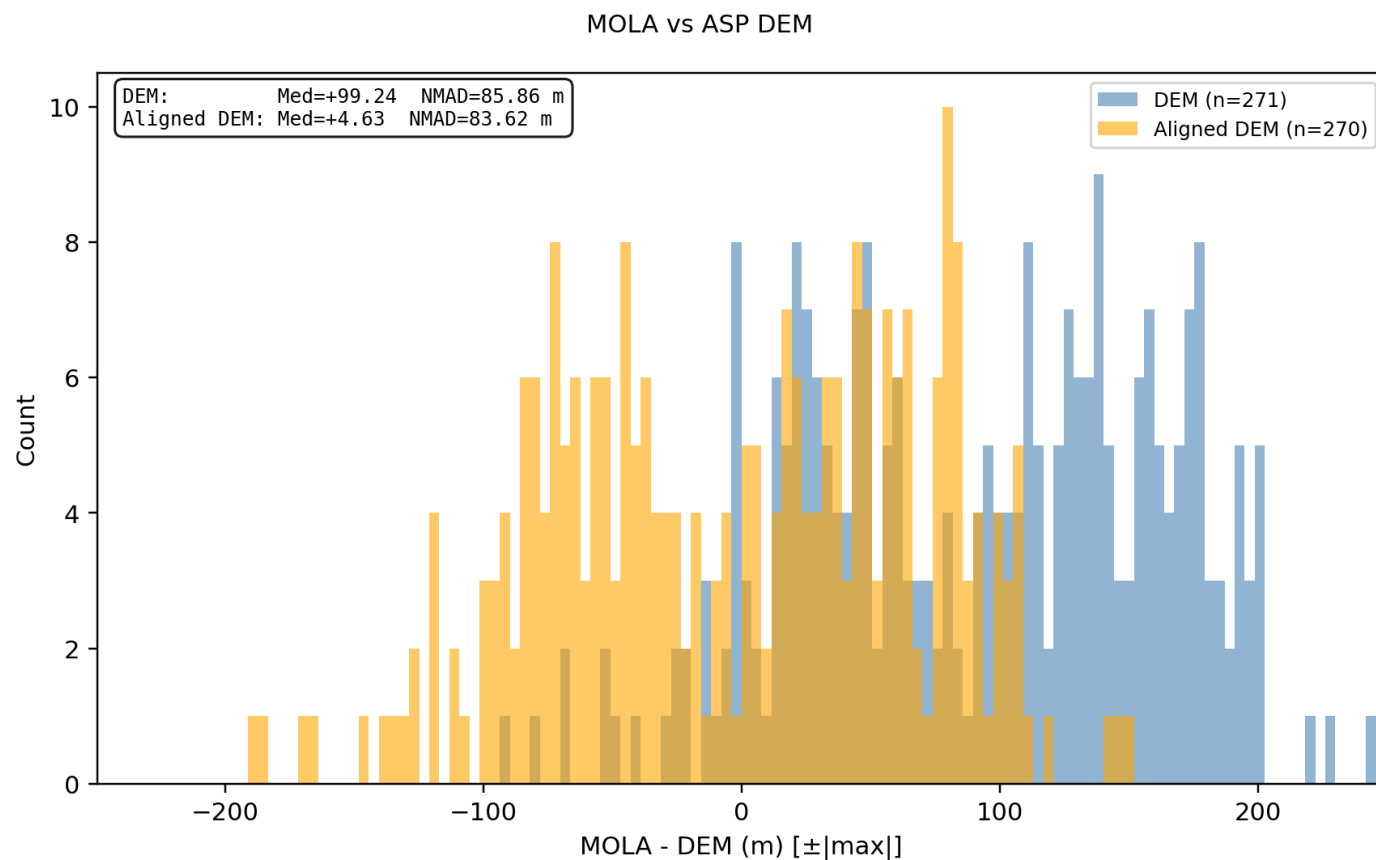


Figure 10: Pre- (steelblue) and post-alignment (orange) distributions of MOLA minus DEM height differences with shared bin edges.